Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 19)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD_{50} for pest / mg kg^{-1}	LD_{50} for ground beetle / mg kg^{-1}	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 1 . 1 Explain what is meant by LD_{50}.

[1 mark]

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0 1 . 2 Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]

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0 1 . 3 Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]

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0 1 . 4 Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

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0 1 . 5 Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

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0 2 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 2 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

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0 2 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 3 Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

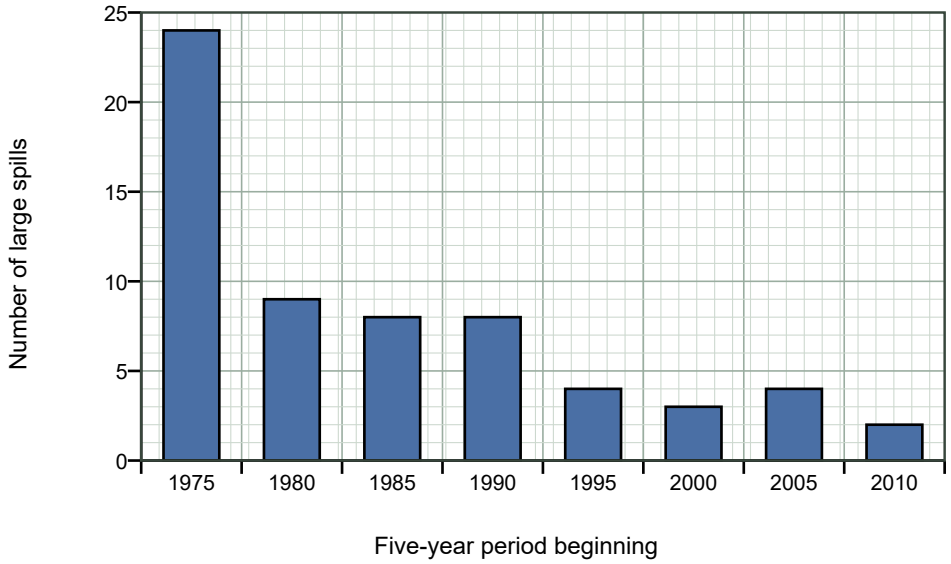


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 3 . 1 Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

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Answer [ ] %

0 3 . 2 Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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0 3 . 3 Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

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0 3 . 4 Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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0 4 Table 1 shows the water quality required for four different uses.

0 4 . 1 Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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0 4 . 2 Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

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0 4 . 3 Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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0 4 . 4 Give **two** methods used to sterilise treated water.

[2 marks]

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0 5 . 1 Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

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0 5 . 2 Explain why recycling is not always energy efficient.

[2 marks]

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0 5 . 3 Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

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High-voltage grid .....

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ICT management of supply and demand .....

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0 5 . 4 Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

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0 6 . 1 Explain how climate change may affect the survival of species. Use named examples.

[6 marks]

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0 6 . 2 Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

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0 6 . 3 Explain why polar regions warming faster than the equator affects the weather in the UK.

[2 marks]

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0 7 . 1 Give **three** sources of mercury pollution.

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0 7 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

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0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 7 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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0 8 Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

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0 8 . 1 Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

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0 8 . 2 Explain why affluence increases per capita energy consumption.

[3 marks]

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0 8 . 3 Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

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0 8 . 4 Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

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0 8 . 5 Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

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0 9 . 1 Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

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0 9 . 2 Explain how a **cyclone separator** works.

[2 marks]

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0 9 . 3 Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

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0 9 . 4 Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

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0 9 . 5 Explain why smaller smoke particles (PM1) are a greater health risk than larger ones.

[2 marks]

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1 0 . 1 Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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Extra space

1 0 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS